-POS tagging is the process of assigning the propriety part of speech tag to each word in a given sentence. Inputs are tokens and the output is part of speech tag for the given tokens.

- A statistical POS Tagger scans tokens and assigns POS tag to each token. POS tagging is used to determine the grammatical value of a word in a given sentence.

-POS Tagging require a training phase and a tagging algorithm, such as Decision Trees, Hidden Markov Models, Support Vector machine and Transformation-based Taggers.

- The Maximum Entropy Model encode features using information about a word, it predicts the optimal tag by calculating the probability of a tag given a context "C".

-Another tag method is the probabilistic model using argmaxP(Y|X), X being the model and Y being the tag sequence

-Generative sequence model decompose probability's using Baye's rule.

-Neural Network based models follow two approaches, Window-based and sentence based. Window-based models analyses features around a specific word in a sentence to perform tagging while sentence based uses the entire sentence as input to perform tagging.

-NER (Named entity recognition) is used to identify and classify the name of a person or organization. from the text to put it in pre-defined categories. The different approaches are dictionary and ruled based approaches, it's simple and fast but it can't deal with name variants and ambiguities. The shallow parsing approach focuses on the internal structures of names. Other approaches are :MITIE, CRF++ and Convolutional neural networks

-BERT output is text representation that understands context well, it can be used for sentence pairing, single sentence classification, Q&A task and single sentence tagging task

-Syntax analysis is the process of analyzing a string of symbols conforming to the rules of formal grammar. It can detect if a sentence is grammatically correct using structural representation of a sentence.

- A parser is an algorithm that calculates the structure of an input string given grammar. It has two fundamental properties: directionality and search strategy. Parsing representation can be done using a tree representation and a list representation.

-Probabilistic language model assigns a probability to a sentence, they are used for spell correction and recognition, but also for summarization and machine translation. To compute P(W) we use P(w1,w2,...,wn) = P(w1)P(w2|w1)P(wn|xw1,...,wn) it's based on chain rule.

-Bigram models use condition on the previous word while the Markov assumption doesn't.

-Neural network-based language model predicts the next word by considering the context, the input word is then converted into a vector.

-Sequence to sequence learning: the input sequence is converted into a hidden state vector by the encoder, the decoder being the output sequence. This process is used in machine translation and can handle problems where the input and output sequence have different lengths.